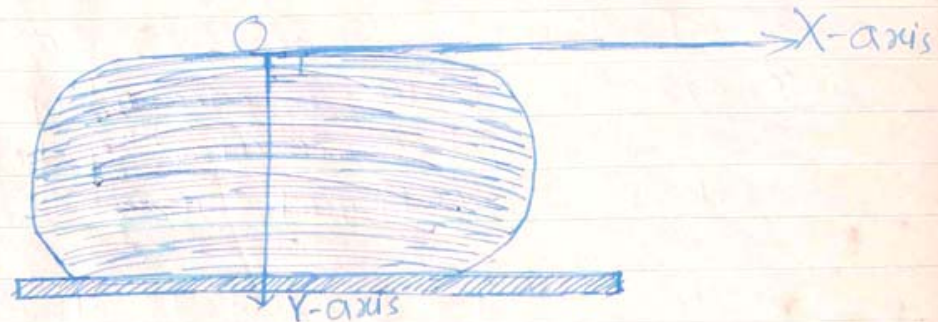




Surface Tension – Sessile Drop

Experimental determination of surface tension of mercury by Quinck's method: Sessile drop:

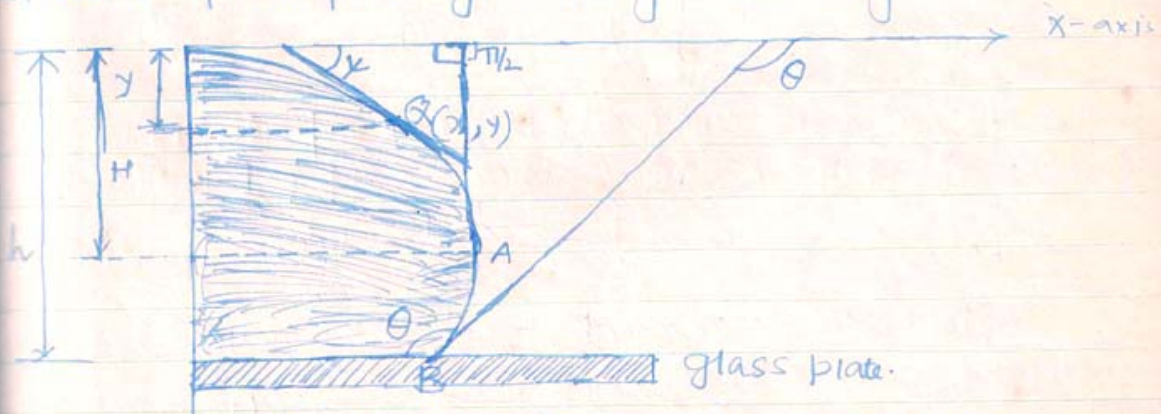
Theory: Let a moderately large drop of mercury be taken on a clean glass plate. Mercury does not wet glass surface.



The highest point 'O' is taken as origin.

A vertical line drawn through 'O' along downward direction is taken as Y-axis. A convenient line perpendicular to Y-axis is taken as X-axis.

Let us consider a section of the mercury drop; by a vertical plane passing through the origin.



Given:- S = Surface tension of mercury.

θ = Angle of contact between mercury & glass.

ρ = density of mercury.

Let us consider any point $Q(x, y)$ on the curved



Vertical Surface of mercury drop.

Let r & R be the principal Radii of curvature at the point Q .

Hence the excess pressure at the point Q :-

$$P = S \left[\frac{1}{r} + \frac{1}{R} \right] \rightarrow (1)$$

Let P_1 & P_2 be the pressure at Q on the concave and convex side of the mercury drop respectively.

Since, pressure at every point, on the same level inside a liquid are same hence

$$P_1 = P_0 + \gamma \rho g \quad \text{where } P_0 = \text{atmospheric pressure}$$

$$P_2 = P_0$$

$$\therefore \text{Excess pressure at } Q : P = P_1 - P_2 = \gamma \rho g \rightarrow (2)$$

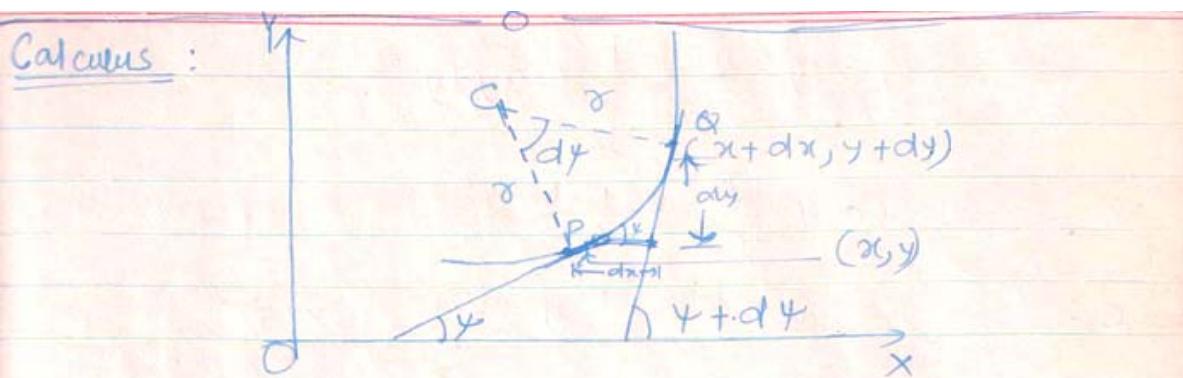
Equating (1) and (2) :-

$$S \left(\frac{1}{r} + \frac{1}{R} \right) = \gamma \rho g \rightarrow (3)$$

Since γ for a mercury drop; One of the two principal radii is very large compared to the other

$$R \gg r \quad \text{or} \quad \frac{1}{R} \ll \frac{1}{r} \quad \text{hence } \frac{1}{R} \text{ can}$$

be neglected compared to $\frac{1}{r}$. Equation (3) reduces to $S \cdot \frac{1}{r} = \gamma \rho g \rightarrow (4)$



Let r = Radius of curvature at P. P & Q are so close; that position P & Q can be assumed to be uniformly bent.

Angle between the two tangents is $d\psi$ hence the angle between the normals drawn to the two tangents is also $d\psi$ i.e. angle between the radii is $d\psi$.

$$d\psi = \frac{d\phi}{r} \quad \text{or} \quad \boxed{\frac{1}{r} = \frac{d\phi}{ds}}$$

$$\boxed{\sin \psi = \frac{dy}{ds}} \quad \boxed{\cos \psi = \frac{dx}{ds}} \quad \boxed{\tan \psi = \frac{dy}{dx}}$$

Let the tangent at Q; makes an angle ψ with the x-axis and s is the arclength; measured along the curve from any fixed point, say origin.

From Calculus; the curvature at Q:

$$\frac{1}{r} = \frac{d\psi}{ds}, \quad \sin \psi = \frac{dy}{ds}, \quad \cos \psi = \frac{dx}{ds}$$

Putting the value in Equation ④

$$s \cdot \frac{d\psi}{ds} = \gamma \rho g \quad \text{or} \quad s \cdot \frac{d\psi}{dy} \cdot \frac{dy}{ds} = \gamma \rho g$$

$$\text{or } s \cdot \sin \psi \cdot \frac{d\psi}{dy} = \gamma \rho g$$

$$\text{or } s \cdot \sin \psi \cdot d\psi = \gamma \rho g \, dy \quad \text{Integrating both sides:}$$



$$\int_0^{\psi} S \cdot \sin \psi \cdot d\psi = \int_0^y \rho g y dy$$

$$\text{or } S [-\cos \psi]_0^{\psi} = \frac{1}{2} \rho g y^2$$

$$\text{or } S [1 - \cos \psi] = \frac{1}{2} y^2 \rho g \longrightarrow (5)$$

As we go from the point Q to B along the curved surface of the mercury drop ψ changes from 0 to θ , acute to obtuse value, hence it must cross $\pi/2$ at some point.

Let A be the point on the curved mercury surface, where the tangent is vertical i.e. $\psi = \pi/2$ & let H be the depth of that point.

$$\text{i.e. } [y = H] \quad \& \quad [\psi = \pi/2] \quad \text{from (5)} : S = \frac{1}{2} H^2 \rho g \longrightarrow (6)$$

At the point B, in contact with the glass
 $[y = h] \quad [\psi = \theta = \text{angle of contact}]$

$$\text{from (5)} : S [1 - \cos \theta] = \frac{1}{2} h^2 \rho g \longrightarrow (7)$$

Using Equation (6) 'S' can be calculated and then using that value of 'S' in (7); angle of contact θ can be calculated.

Correction: In deriving Equation (7) $\frac{1}{R}$ was neglected compared to $\frac{1}{r}$; hence a correction term is applied. It can be shown that the corrected formula is

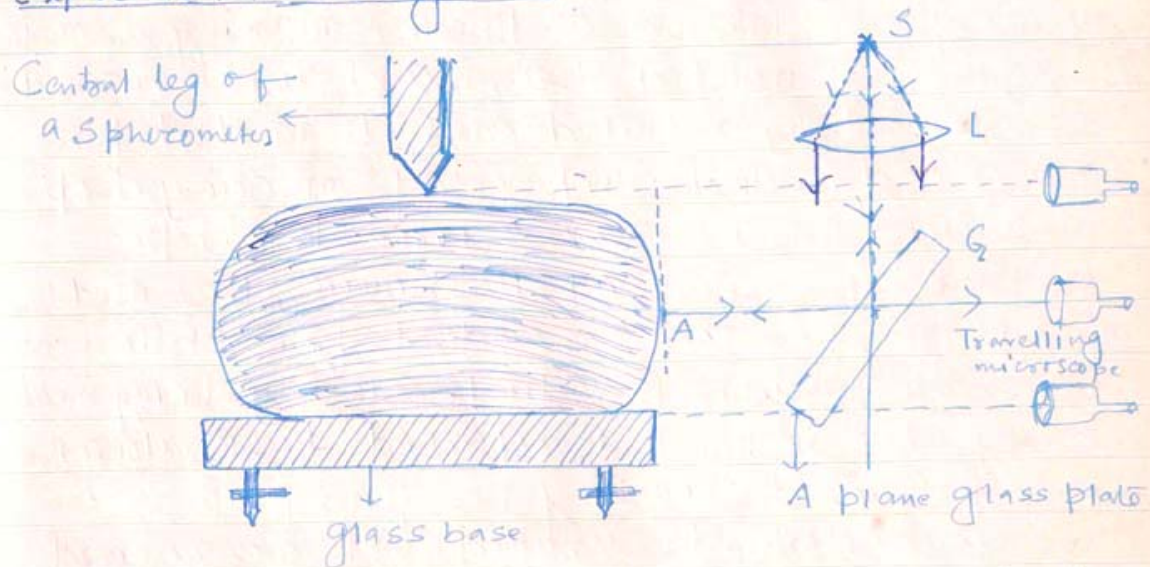
$$S [1 - \cos \theta] = \frac{1}{2} h^2 \rho g \left[1 - \frac{0.61 h}{c} \right] \longrightarrow (8)$$



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Where c is the diameter of the circular horizontal flat surface of the mercury drop.

Experimental arrangement:



A glass base providing with levelling screws is thoroughly cleaned first with acid, then with alkali and finally with distilled water to make it free from any oily or greasy substance. The glass base is levelled properly. The mercury is also cleaned by running it through a capillary tube. A moderately large drop of mercury about 8 to 10 cm in diameter is taken on the glass base. The central leg of a spherometer is first made to touch the highest point on the mercury surface, which acts as origin.



A source of light 'S' is kept at the focus of a convex lens L; and the horizontal beam of parallel



Rays issuing from the lens is incident on a plane glass plate G . Kept with its plane vertical, inclined with the axis of lens at an angle of 45° . The horizontal beam is partially reflected by the glass plate in a direction making an angle $\pi/2$ with the incident beam and this horizontal beam of partially reflected rays is made to fall on a curved vertical surface of mercury drop which acts as a convex mirror. By proper adjustment of height of the source, lens and the glass plate; the ray is made to fall at the point where the tangent is vertical. The ray being incident normally at that point is reflected back along the same path and reaching the glass plate, the rays are partially transmitted and are received by the travelling microscope. A bright image of the source S is seen at the cross-wire of the eye-piece. The microscope is then raised up and is focussed at the tip of the central leg of the spherometer and the reading is noted. The microscope is then brought down and is focussed at the upper edge of the glass base and the reading is noted. From these three readings ' h ' & ' H ' can be calculated. The diameter of the circular flat surface is measured by a pair of dividers.

Thus knowing ' H ', ' h ' & ' C ', ' S ' & ' θ ' can be calculated.